

Which Pass Rate Answers Which Question?

Unit 2 · Topic 2.2 · TODAY'S PROBLEM

Suppose a school records exam outcomes for 300 students. Students chose for themselves whether to use the tutoring center.

Mara: *The pass rate is 75%.*

Noah: *The pass rate is 88%.*

Elena: *Tutoring raises the pass rate by 13 percentage points because $88\% - 75\% = 13$ percentage points.*

Exam outcomes for 300 students

Center?	Passed	Did not pass	Total
Used	66	9	75
Did not	159	66	225
Total	225	75	300

FIRST TAKE (5 min, on your sheet)

Can 75% and 88% both be correct pass rates? State what group you think each percentage describes.

- DOK 1** (a) Identify 75% and 88% as marginal or conditional relative frequencies. For the conditional frequency, name the group on which it is conditioned.
- DOK 2** (b) Verify both percentages from the table. Then compute the pass rate among students who did not use the center and the percentage-point gap between the two user groups.
- DOK 3** (c) Explain which question Mara's 75% answers and which question Noah's 88% answers. Decide whether tutoring use and exam outcome appear associated, citing the two conditional pass rates that settle it. Then evaluate Elena's 13-point claim: state what two rates she compared, what comparison directly measures the group gap, and why the word *raises* is not warranted when students chose tutoring.

Video + follow-along: [u2_lesson3_live.html](https://www.illustrativemathematics.org/HS/index.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

